



(∞, ∞) -categories w/adjoints
are a bit like spaces

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Simons Collaboration on Global
Categorical Symmetries

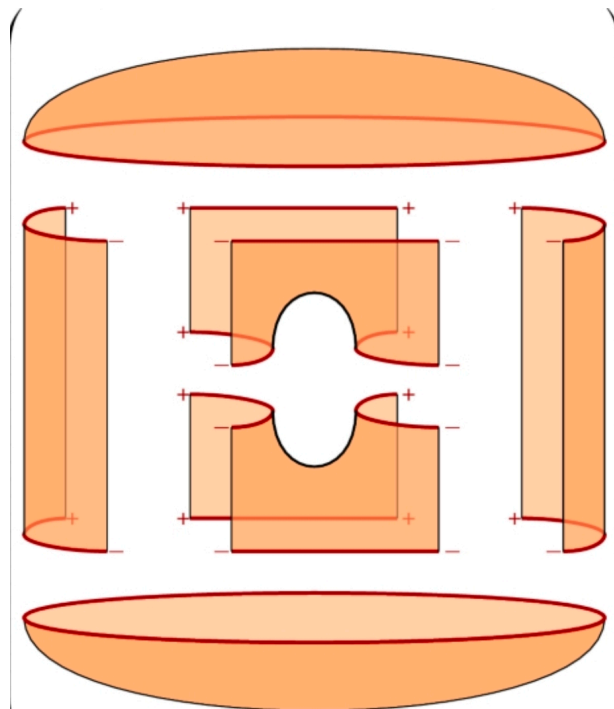
∞ -CATS ∞ -CATS
w/ADJOINTS SPACES



invertibility

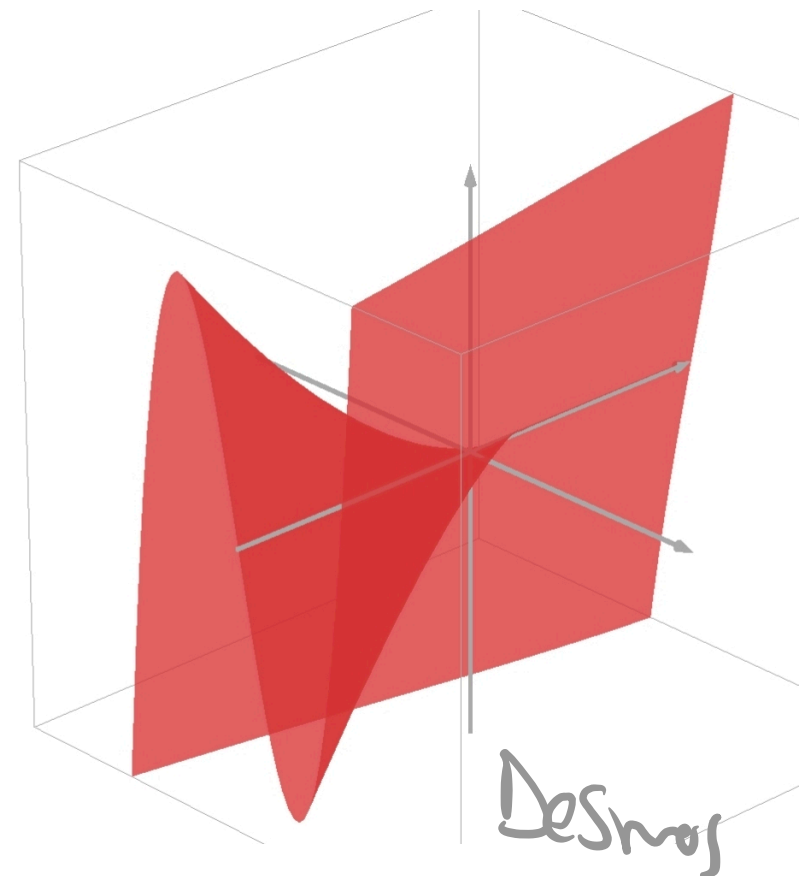
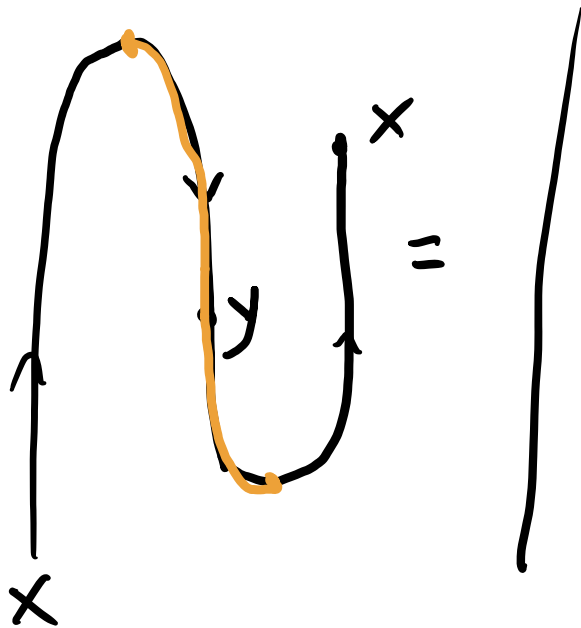
TODAY'S : the Cobordism Hypothesis is
 GUIDING : AKA
 THREAD : a universal property of bordism

Thm (C.H.) $\text{Bord}_n^{\text{fr}}$ is the free symmetric (∞, n) -category
 w/ duals.



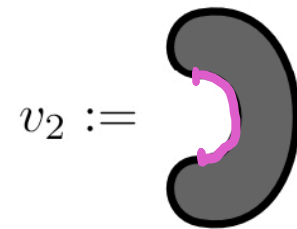
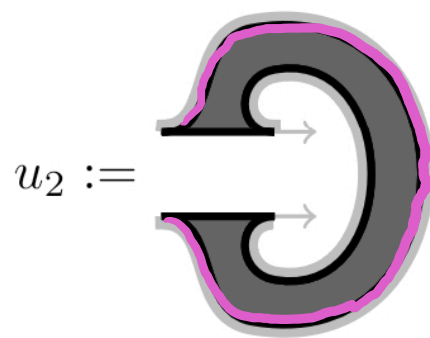
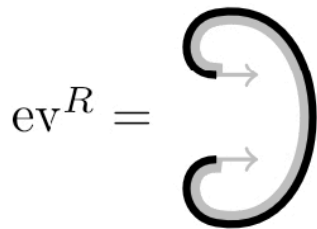
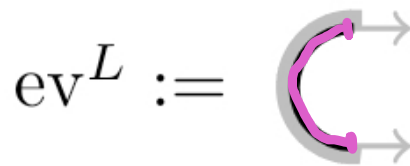
= a 2-morphism
 in Bord_2

$\dim = 1$
(Bord₁)



$\dim = 2 \rightarrow 2\text{-dualizability}$
 Bord_2

$\downarrow$
 morphisms have
 adjoints



Douglas-Schumacher-Pries-Snyder

An n -category is ^{roughly} the following data:

→ a set of objects

→ a set of morphisms btw each pair of objects

→ " 2 -morphisms " of morphisms

⋮
→ a set of n -morphisms btw each pair of $(n-1)$ -morphisms

An (∞, n) -category is ^{roughly} the following data:

→ a ~~space~~ set of objects

→ a ~~space~~ set of morphisms btw each pair of objects

→ " 2-morphisms " of morphisms

→ a ~~space~~ set of n -morphisms btw each pair of $(n-1)$ -morphisms

① SPACES & ∞ -CATEGORIES

Principle: our building blocks
are homotopy types.

→ "HOMOTOPY HYPOTHESIS"

→ Kan complex
" ∞ -groupoid
" anima.

Def. a category is:

- a space of obj's
- a space of morphisms btw each pair of obj's

TIMELINE:

not historically accurate

1900 - π_1
- H_n
1930 - π_n, H^n
1940
1945 - CATEGORIES
1950 - BG
1958 - SPECTRA
1960 - MODEL CATS
1970 - E_n -OPERADS
1973 - WEAK KAN COMPLEXES
1980

ALGEBRAIC
TOPOLOGY

HOMOTOPY
THEORY

CATEGORY
THEORY

1985 - BRAIDED CATS

1990 -

1995 - COBORDISM
HYPOTHESIS

2000 -

∞ -CATEGORIES

2006 - HTT

PROOF OF C.H.

2009 - HA

2010 -

MODELS OF
 n -CATS

2020 - ...

HOMOTOP.
CAT.
THEORY

$$n\text{Cat} = \begin{cases} \text{a space of obj's} \rightarrow \text{Ob} \\ \text{an } (n-1)\text{-cat btw each } x, y \in \text{Ob} \end{cases}$$

Def

$$\infty\text{Cat} = \varprojlim (\dots \rightarrow 2\text{Cat} \rightarrow 1\text{Cat} \rightarrow 0\text{Cat})$$

$\parallel$
 $\int$

DOING HOMOTOPY

W/ ∞ -CATEGORIES

→ products

→ wedge

→ disks

→ loops

→ spheres

→ suspension

see recent work of Gepner, Hove & Masuda

② SPECTRA & MONOIDAL CATEGORIES

$$\begin{array}{ccccc} A & = & \Omega(BA) & = & \Omega^2(B^2A) \\ \uparrow & & \uparrow & & \uparrow \\ \text{Mon} & & \text{Cat}_* & & 2\text{Cat}_* \end{array}$$

② SPECTRA & MONOIDAL CATEGORIES

E_2 -spaces (e.g. $\bigcup B \text{Braid}_n$)

$$A = \Omega(BA) = \Omega^2(B^2A)$$

~~$\mathcal{M}$~~
 $\text{Mon}(S)$

$\mathcal{C}at_*$

$2\mathcal{C}at_*$

② SPECTRA & MONOIDAL CATEGORIES

$$\begin{array}{ccccccc} A = \Omega(BA) = \Omega^2(B^2A) = \Omega^3(B^3A) = \dots \\ \uparrow \quad \quad \uparrow \quad \quad \quad \uparrow \\ \text{Mon(S)} \quad \text{Cat}_* \quad \quad 2\text{Cat}_* \end{array}$$



DEF (Lima)₁₉₅₈: Ω -spectrum

SYMMETRIC
MONOIDAL

N-CATEGORIES

ARE CATEGORICAL
SPECTRA!

DOING HOMOTOPY W/ CATEGORICAL SPECTRA

→ SMASH

→ DOLD-THOM

→ DOLD-KAN?

→ BROWN REPRESENTABILITY

→ SSEQS?

→ ~~WHITEHEAD~~

→ CHAIN HOMOLOGY?

→ CELLULAR HOMOLOGY?

Proof of the CH (Lurie)

induction: $\left\{ \begin{array}{l} \text{base} = \text{Bord}_1 \\ \text{inductive step: } \text{Bord}_{n-1} \hookrightarrow \text{Bord}_n \end{array} \right.$

$\textcircled{i} \text{Bord}_n = \text{Bord}_n$ ^{+ framed function}

filter by
Morse index

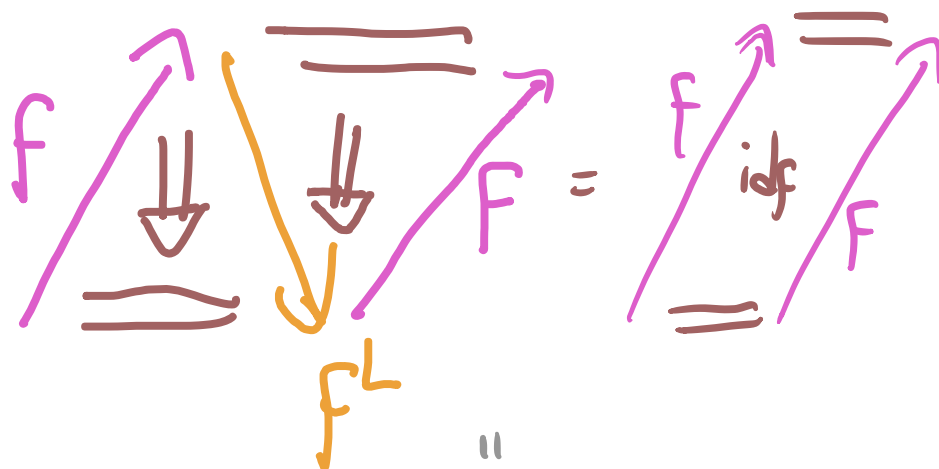
$\textcircled{ii} \text{Bord}_{n-1} \hookrightarrow F_1 \hookrightarrow F_2 \hookrightarrow \dots \hookrightarrow F_n = \text{Bord}_n$ ^{+ framed function}

③ CATEGORIES W/ DUALS

& WHAT I DO W/MY TIME

→ CATS
arrows

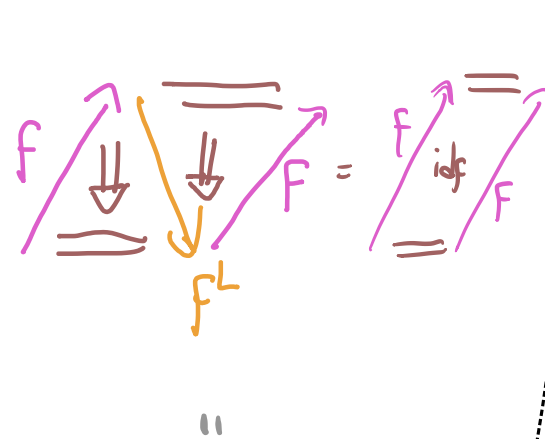
adjoints



$\sim$

→ SPACES

isos



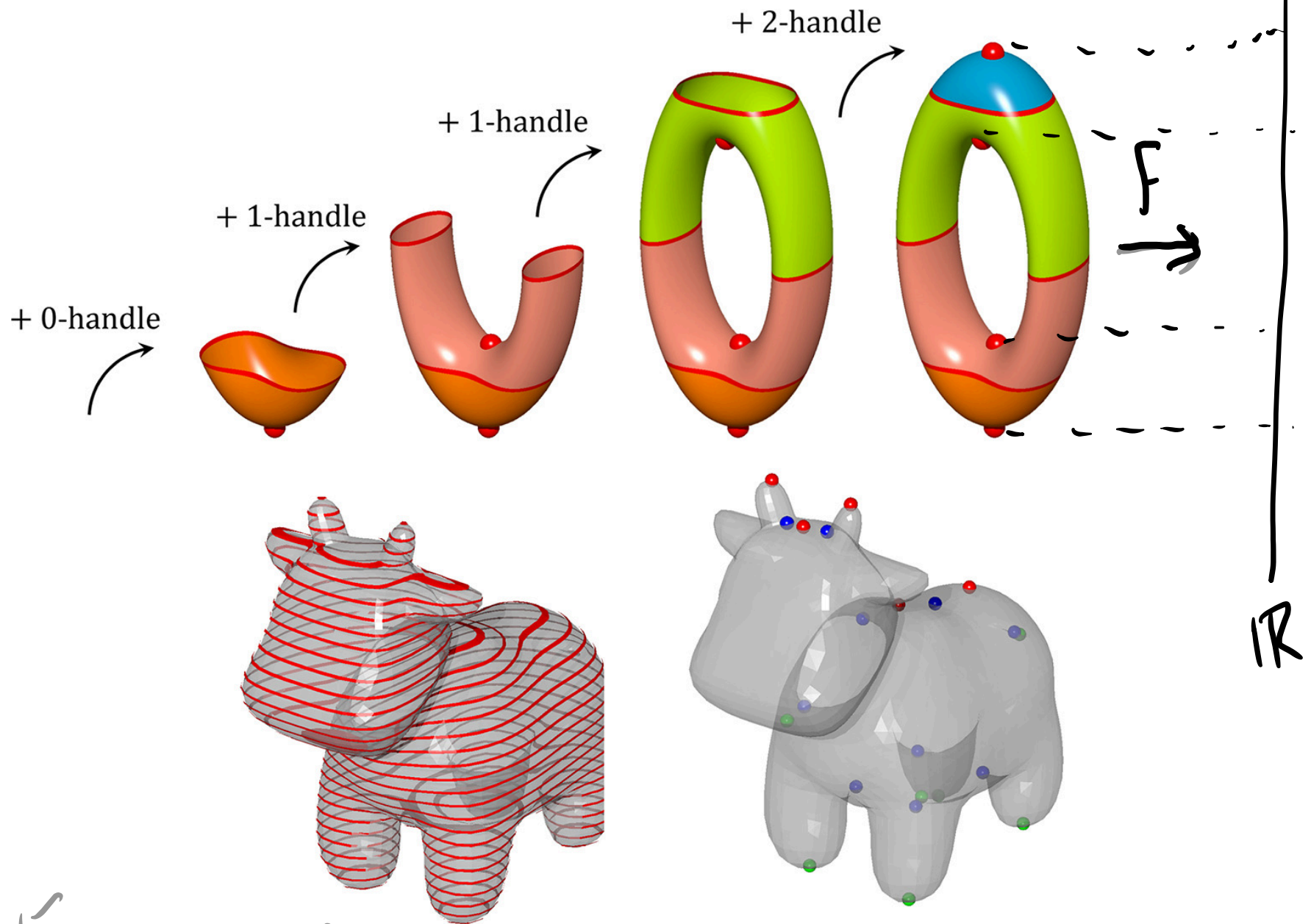
$\sim$

Def An n -category has adjoints if every $(< n)$ -morphism has both adjoints.

SPACES	∞ -CATS	n -CATS W/ADJ
$S' = B\mathbb{Z}$	$\vec{S}' = B\mathbb{N}$	$Bord_1$
$\$$	$FinSet_*$	$Bord_n^F$
$ Conf_{\mathbb{R}^k}(n) = \mathcal{P}^k S^k$	$Conf_{\mathbb{R}^k}(n) = \mathcal{P}^k \vec{S}^k$	$Tang_{n,k}$
∞ -loop spaces	$Sym Mon Cat$	$Sym Mon Cat$ w/ duals

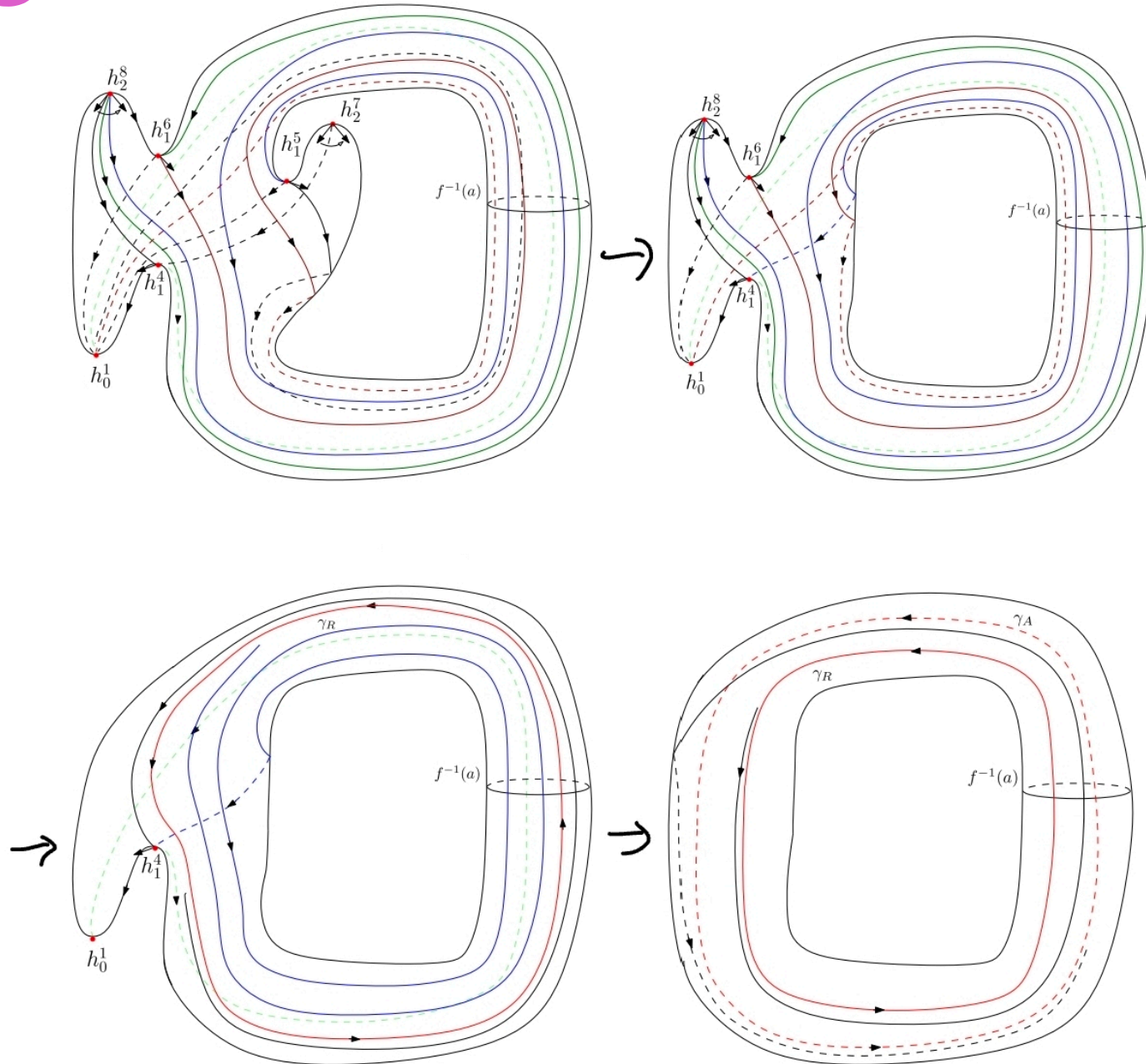
↪ $Sym Mon Cat$ w/ inverses

Why $\text{Bord}_n = \text{Bord}_n^{\text{Framed}}$



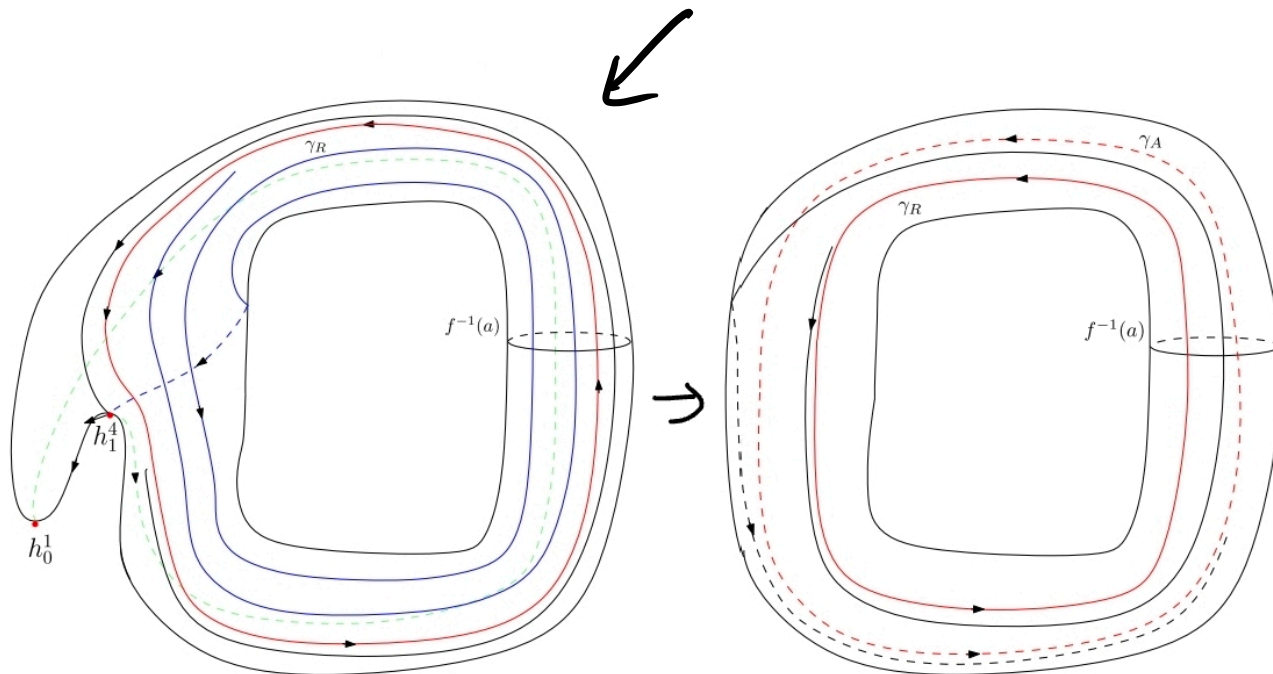
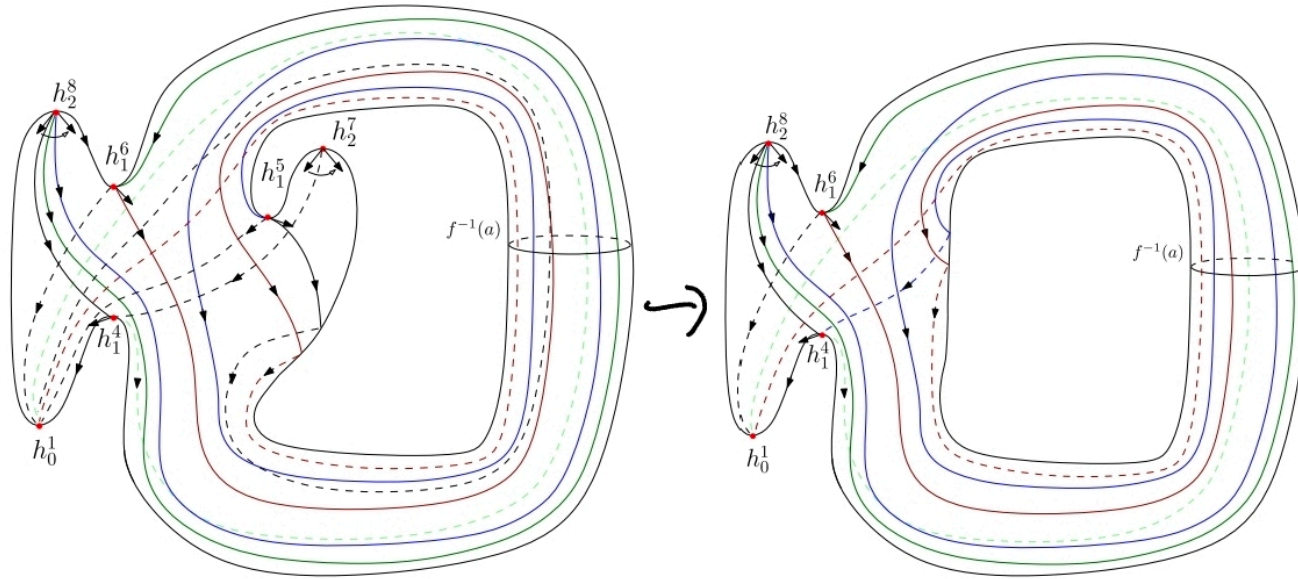
↙
Pereira-Corcho et al, 2023

Why $\text{Bord}_n = \text{Bord}_n^{\text{framed}}$ function



Lima-Neto-Rezende-Silveira (2016)

Why $\text{Bord}_n = \text{Bord}_n^{\text{framed function}}$ $\rightarrow$ Cerf + some data



Lima-Neto-Rezende-Silveira (2016)

HYPOTHESIS: the "homotopy theory" of ∞ -cats w/ adjoints is not that bad

SPACES	∞ -CATS	n -CATS W/ADJ
$S' = B\mathbb{Z}$	$\vec{S}' = B\mathbb{N}$	$Bord_1$
$\$$	$FinSet_*$	$Bord_n^F$
$ Conf_{\mathbb{R}^k}(\bullet) = \mathcal{S}^k \mathcal{S}^k$	$Conf_{\mathbb{R}^k}(\bullet) = \mathcal{S}^k \vec{\mathcal{S}}^k$	$Tang_{n,k}$
∞ -loop spaces	$SymMonCat$	$SymMonCat$ v/ duals
$\hookrightarrow SymMonCat$ w/ inverses		

TO-DOs

- ① Translate the proof of the C.H. to the language of cat.spectra w/adjoints.
What do you learn?
- ② [Postnikov Theory for n -cats w/adjoints?
↳ my (in progress) thesis
- ③ Can we deepen the relation to singularity theory?

BONUS: degenerate singularities & coherence in cats

Lurie CH proof, section 3.4:

the generic behavior of several parameter families of functions on B . This is feasible for small numbers of parameters (and in fact this is sufficient for our purposes: we can use the methods of §3.5 to reduce to the problem of understanding the $(2, 1)$ -categories $\tau_{\leq 2} \mathcal{B}(M)$) using Thom's theory of *catastrophes*. However, it will be more convenient to address the issue in a different way, using Igusa's theory of *framed functions*.

The first seven elementary catastrophes

Name	Symbol	Crank	Codimension	Germ	Unfolding terms
Fold	A_2	1	1	$x^3 + y^2$	x
Cusp	$A_3^\pm$	1	2	$\pm x^4 + y^2$	x^2, y
Swallowtail	A_4	1	3	$x^5 + y^2$	x^3, x^2, x
Butterfly	$A_5^\pm$	1	4	$\pm x^6 + y^2$	x^4, x^3, x^2, x
Hyperbolic umbilic	D_4^+	2	3	$x^3 + xy^2$	$x^2 + y^2, x, y$
Elliptic umbilic	D_4^-	2	3	$x^3 - xy^2$	$x^2 + y^2, x, y$
Parabolic umbilic	D_5	2	4	$x^4 - xy^2$	x^2, y^2, x, y

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Swallowtail	A_4	1	3	$x^5 + y^2$	x^3, x^2, x
Butterfly	$A_5^\pm$	1			
Hyperbolic umbilic	D_4^+	2			
Elliptic umbilic	D_4^-	2			
Parabolic umbilic	D_5	2			

swallowtail
equation
↑

